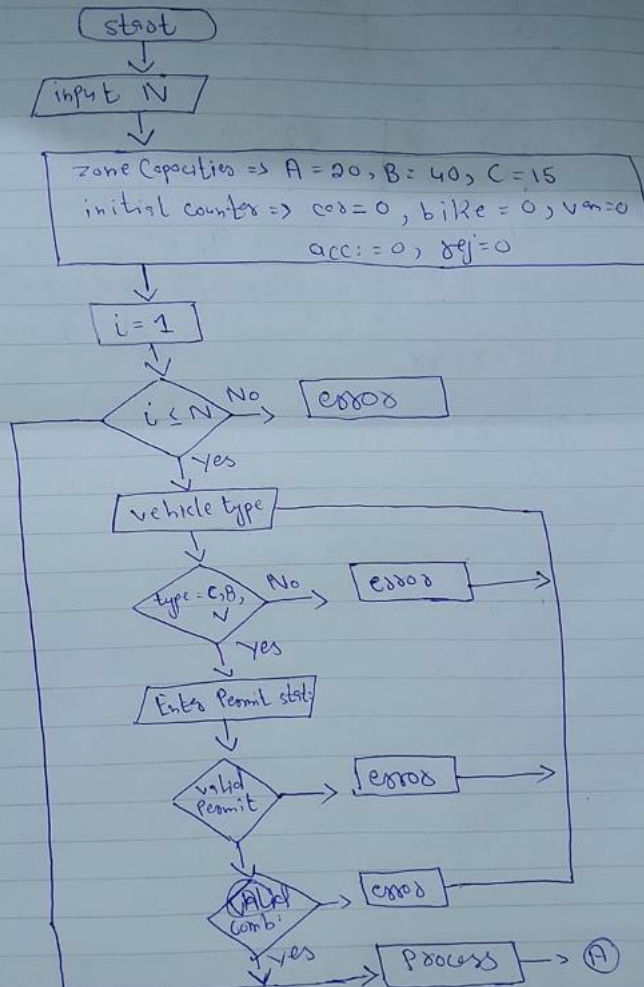
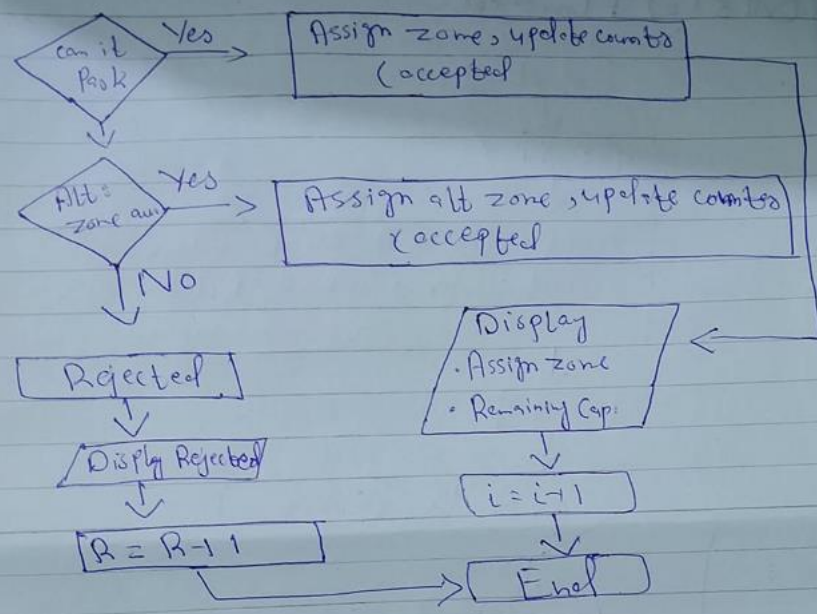


Q# 5 (b) Flowchart





Pseudocode

```
1. start
2. initialize zone capacities
   Zone A = 20, B = 40, C = 15
   set count: car = 0, bike = 0, van = 0
   Acc = 0, Rej = 0
3. Input N
4. For i = 1
   Repeat
     input vehicle type
   Repeat
     input category
   Repeat
     input Permit
   if Permit = N
     ask if emergency vehicle
   check eligibility
   if eligibility then
     check zone Capacity and assign
     • update zone occupancy
     • update counter
     • acc = acc + 1
     • Display zone
   else
     Rej = rej + 1
     Display rej
   End if
End For
```

5. Display Parking Summary
 6. if total occupancy = total cap:
 Display "Parking is full"
 else
 Display "Parking is not full"
 7. End

PAC - Chart :

Step	Description	Input	Output
1. Input N	Enter no. of veh:	N	-
2. validate input	Check vehicle type category, permit, emergency	type, category, permit, Emerg	Valid / Invalid
3. Checks eligibility	Apply rules	veh + cate	Eligible / not eligible
4. Assign zone	Check capacity, apply alternatives	Zone status	Zone ass: ok / reject
5. update counts	Update occupancy and counters	Zone, type	Update value
6. Loop	Repeat for next vehicle	i	-
7. Summary	show final results	All data	Summary report

IPO - chart :

Input	Process	Output
no. of vehicle to be processed	initialize counters and zone occupancies	For each veh: • Assign zone and remaining cap. • Reg. reason
For each veh: - veh type - user cate - Permit - Emergency vehicle Zone Cap: - A = 20 - B = 40 - C = 15	- validate all input (V.T, Cat, Permit, Emerg.) - For each vehicle, det. eligibility using rules - Check zone cap: and assign zone - Update occupied or remaining space - maintain counters - After all vehicle, generate summary and check highest occupancy.	- Parking Summary - Total Processed - Acc / Reg - Cat, bike, van - Final occ / remain. cap. of each zone - Zone with highest occupancy - whether Campus is full